



LA MENARA — THE 12 REMARKABLE RELAYS

37 ISI Worked Examples — The Integrated Spreadsheet Companion

Approximately 3 worked examples per relay — covering all 12 civilisational relays

Ir. Nigel T. Dearden BEng(Hons) CEng MICE MHKIE MCIWEM C.WEM RPE(CSB)

Chartered Civil & Structural Engineer | 36 Years Professional Practice (1989–2026)

*From Calories to Consciousness: A Civil Engineer's Perspective and Guide —
An Infrastructure Odyssey*



iAAi COUNTERFORCE v68 — Infrastructure Academy — June 2026

◆ The line is not safe until the system is tested. ◆

THE FIRST CAUSE: INFRASTRUCTURE EDUCATION IS FRAGMENTED, ABSTRACT, AND DISCONNECTED FROM REAL-WORLD IMPACT



FRAGMENTED LEARNING

70%

of Curriculums Lack
Interdisciplinary Integration

[1] Hong (2025), Nature

45%

of Graduates report limited
exposure to system-level thinking

[3] AAC&U (2023)



ABSTRACT CONCEPTS

65%

of Students Feel Unprepared
for Practical Application

[2] Hult / NACE (2025)

80%

of Employers see a skills gap
in project readiness

[4] Ibec (2026) / McKinsey (2022)



DISCONNECTED FROM IMPACT

Only 30%

of Programs integrate community
or environmental impact metrics

[5] Leifler & Dahlin (2020), IJSHE

90%

of Infrastructure Projects face delays
due to practical knowledge gaps

[6] Flyvbjerg (2014), PMJ



URGENT NEED: Bridging the Gap for Future-Ready Infrastructure Professionals

THE PLANETARY ENGINE — PROLOGUE

Earth as Fusion Reactor with Magnetic Core

The Cosmic Substrate

- 4.5 billion years of geological infrastructure
- Magnetic core generates the protective field
- Solar fusion powers all biological energy
- From this engine, all 12 relays emerge



La Menara Opens Here

- The Prologue before R01 (Fire)
- Earth is not passive — it is the first machine
- Calories begin as stellar fusion
- Consciousness ends as the Fractal Connector



From Planetary Engine to Fractal Connector: 12 Civilisational Relays — The Complete Journey



THE 12 CIVILISATIONAL RELAYS — FROM FIRE TO HUMAN NODES

The Relay Race That La Menara Narrates — 12,000 Years of Infrastructure Evolution

1



R01 FIRE

pre-10,000 BCE

The First Spark of Intent

2



R02 TREE

pre-10,000 BCE

The First Canopy

3



R03 RIVER

8,000 BCE

Cradles of Continuity

4



R04 HORSE

4,000 BCE

Outrider and Engine of Mobility

5



R05 ROADS

500 BCE

Infrastructure of Velocity

6



R06 SHIPS

500 CE

Oceans as Superhighways

7



R07 LOOM

1780 CE

The Binary Birth

8



R08 RAIL

1830 CE

The Iron Pulse

9



R09 ENGINE

1850 CE

The Relentless Fire

10



R10 AAA TRIAD

1900 CE

Automobile, Aviation & Airwaves

11



R11 ORBIT

1960 CE

The Celestial Relay

12



R12 HUMAN NODES

2000 CE

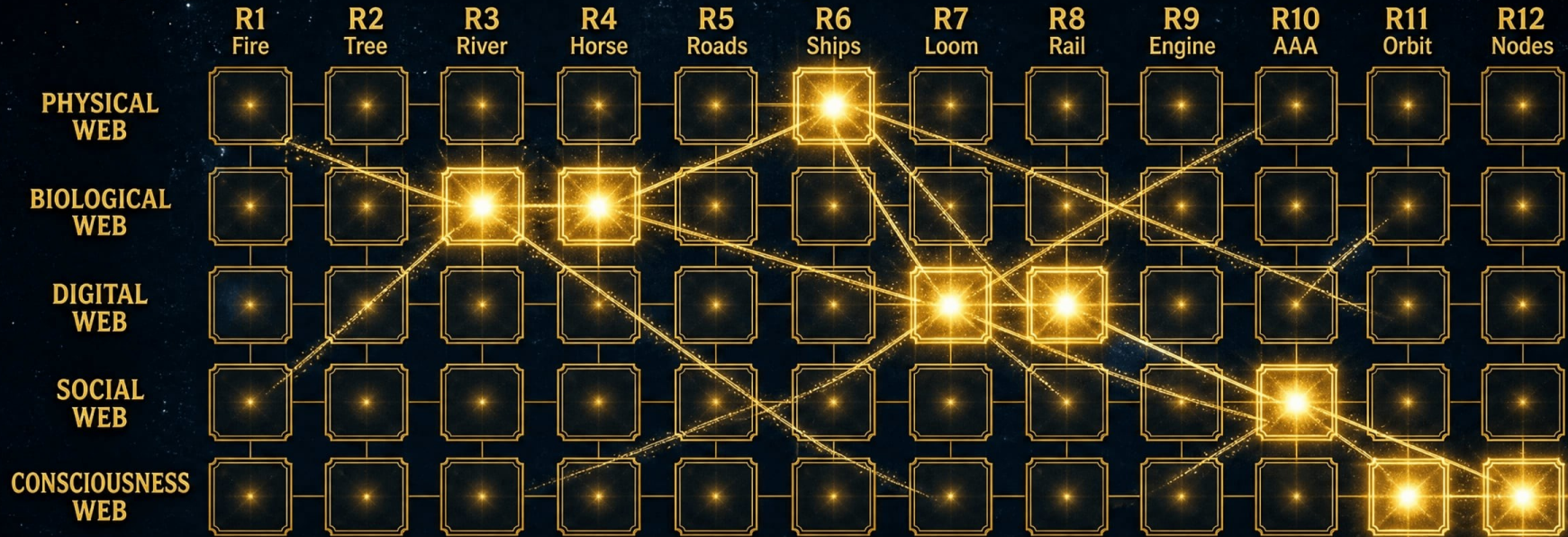
The Biological Relay



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THE DEARDEN FIELD

5 Great Webs × 12 Relays = 60 Nodes of Discovery



The 37 worked examples illuminate 37 of these 60 nodes.
This is not a timeline. It is a fabric. Every relay operates across all five webs.



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THE ISI FORMULA – 0.1 to 10.0 CONTINUOUS SCALE

Excel Sheet 5: Formula Reference

INPUT VARIABLES

- A** (Ambition) = 0.1–10.0
Scale of intent and reach
- P** (Persistence) = 0.1–10.0
Duration and continuity of effort
- β** (Beta / Drag) = 0.1–10.0
Aggregate resistance from 4Cs:
Conflict, Climate, Contagion, Cost
- λ** (Lambda / Ethical Load) = 0.1–10.0
Aggregate ethical friction:
Doctrine, Grace, Interpretation

FORMULAE

$$R_{add} = \beta + \lambda$$
$$R_{mult} = \beta \times \lambda$$
$$C_{add} = \frac{A \times P}{R_{add}}$$
$$C_{mult} = \frac{A \times P}{R_{mult}}$$
$$\rho = \frac{C_{add}}{C_{mult}} \quad (\text{Sensitivity Ratio})$$

TRAFFIC LIGHT THRESHOLDS

-  $\rho < 2.0$
System is robust
-  $2.0 \leq \rho < 5.0$
System is sensitive
-  $\rho \geq 5.0$
System is fragile



$$ISI_1 \otimes ISI_2 \otimes ISI_3 = ISIs$$

Infrastructure Significance Indicator — The Civilisational Output

$$ISI = \sum \frac{(A_i \times P_i)}{(N \times \beta)}$$

i = each node (person) in the network

N = total number of nodes

A = Ambition of each node (0.1–10.0)

P = Persistence of each node (0.1–10.0)

β = Aggregate drag resistance (4Cs: Conflict, Climate, Contagion, Cost)

ISI₁

SUSTAINABILITY

UN SDG Alignment

Does this infrastructure serve the 17 Sustainable Development Goals?

First-order metric — not optional, not an add-on

ISI₂

SURVIVAL

Clock Mode

Does this infrastructure persist through temporal cycles?

Expansion → Consolidation →
Collapse → Renewal

ISI₃

\$IGNIFICANCE

Innate Value

Does this infrastructure carry civilisational worth beyond immediate function?

Legacy quotient — what echoes in eternity



$\otimes$ = Modus Tecton Tensor — the binding operation
 $ISIs = ISI_1 \otimes ISI_2 \otimes ISI_3$ = Infrastructure Significance Indicator
 Three dimensions woven into one civilisational verdict

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$$IQ \otimes EQ \otimes CQ = HQ$$

The Haptic (Holistic/wHole) Quotient — The Individual Output

$$IQ \otimes EQ \otimes CQ = HQ$$

- I = Intelligence — cognitive capacity (what you know)
- E = Emotional — relational capacity (how you connect)
- C = Consciousness — awareness quotient (how you perceive the system)
- H = Haptic — Holistic/wHole Quotient (the complete human metric)
- $\otimes$ = Modus Tecton — the same tensor that binds $ISI_{1/2/3}$

INDIVIDUAL SCALE

$$HQ = IQ \otimes EQ \otimes CQ$$

Measures the single node (person)

What is your capacity to contribute to infrastructure?

HICE = Human Infrastructure Capital Efficiency

The individual metric

FC13
=
THE
DOCKING
POINT

iCU Nexus: $0 + 1 =$ System Node
Where individual meets civilisation
The Fractal Connector binds HQ to ISIs

CIVILISATIONAL SCALE

$$ISIs = ISI_1 \otimes ISI_2 \otimes ISI_3$$

Measures the network (civilisation)

What is the system's capacity to sustain,
survive, and signify?

ISI = Infrastructure Survival Index

The civilisational metric

CQ is not innate — it is created through education, training, and deliberate practice.

This is why iAAi exists. This is what the Infrastructure Academy builds.

iAAi

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ISI MASTER DATABASE — 37 WORKED EXAMPLES

Excel Sheet 1:
ISI Master Database

The 25 Columns



IDENTITY: No., Name, Date, Relay (R01-R12), Zone, Mode



INPUTS: A (Ambition 0.1-10.0), P (Persistence 0.1-10.0)



RESISTANCE: β (Aggregate Drag 0.1-10.0),
 λ (Aggregate Ethical 0.1-10.0)



FORMULAE: R_{add} , R_{mult} , C_{add} , C_{mult} ,
 ρ (Sensitivity Ratio)



CLASSIFICATION: Traffic Light (GREEN/AMBER/RED),
ISI_{1/2/3}



CONTEXT: Initiator, Scholar, Surplus Type,
Collapse Vector

How to Read a Row



Example: Indus Valley (2600 BCE)



Relay: R03 (River) | **Zone:** East | **Mode:** E



A = 7.5 | **P** = 8.0 | **β** = 3.2 | **λ** = 1.5



ρ = 1.62 → **GREEN**



Initiator: Harappan civilisation



Scholar: Sima Qian

Chronological Ordering Principle

All 37 examples flow in strict time sequence: 1.5M BCE → 2026+
The relay race unfolds as it actually happened — not grouped by category.



Ancient



Classical



Medieval



Industrial



Modern



Future



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RELAY COVERAGE MAP — DISTRIBUTION ACROSS 12 RELAYS

Excel Sheet 4:
Relay Coverage Map

37 Worked Examples — Approximately 3 per Relay on Average



R01 FIRE

2

EXAMPLES
pre-10,000 BCE



R02 TREE

2

EXAMPLES
pre-10,000 BCE



R03 RIVER

4

EXAMPLES
8,000 BCE



R04 HORSE

3

EXAMPLES
4,000 BCE



R05 ROADS

4

EXAMPLES
500 BCE



R06 SHIPS

3

EXAMPLES
500 CE



R07 LOOM

2

EXAMPLES
1780 CE



R08 RAIL

3

EXAMPLES
1830 CE



R09 ENGINE

4

EXAMPLES
1850 CE



R10 AAA TRIAD

4

EXAMPLES
1900 CE



R11 ORBIT

3

EXAMPLES
1960 CE



R12 HUMAN NODES

3

EXAMPLES
2000 CE



Total: **37** | Average: **3.1** per relay | Range: **2-4** | All relays covered | **6** Zones | **4** Modes



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SCHOLAR MAPPING — 13 VOICES ACROSS 3,500 YEARS

Excel Sheet 6:
Scholar Mapping

TIER 1: THE 7 SCHOLARS — Foundational Wisdom (~500 BCE – 2026)



HOMER

The Poet
(Narrative)



CONFUCIUS

The Ethics Sage
(Governance)



SUN TZU

The Strategist
(Positioning)



ARISTOTLE

The Philosopher
(Logic)



SIMA QIAN

The Historian
(Record)



MARCO POLO

The Explorer
(Connection)



DEARDEN

The System Builder
(7th Voice)

FEYNMAN (1985) — The Hinge Scholar — Bridging Diagnosis to Application

TIER 2: THE 6 DIAGNOSTICIANS — 1943–1989



MASLOW

Human Potential



PÓLYA

Problem Solving



TURING

Computation



DE BONO

Lateral Thinking



HAWKING

Cosmos



BERNERS-LEE

The Web



Each scholar validates specific relays. The spreadsheet maps which scholar anchors which worked example.



CHRONOLOGICAL FLOW GUIDE — ERA ACTIVATION SEQUENCE

Excel Sheet 7:
Chronological Flow Guide

ERA	TIME RANGE	ACTIVE RELAYS	KEY TRANSITION	EXAMPLES
 ERA 1 — DEEP PREHISTORY	1.5M– 10,000 BCE	R01 (Fire), R02 (Tree)	<i>Control of energy → first material construction</i>	2
 ERA 2 — ANCIENT CIVILISATIONS	10,000– 1000 BCE	R03 (River), R04 (Horse)	<i>Settlement → mobility networks</i>	7
 ERA 3 — CLASSICAL EMPIRES	1000 BCE– 500 CE	R05 (Roads), R06 (Ships)	<i>Velocity infrastructure → oceanic reach</i>	6
 ERA 4 — MIDIEVAL–EARLY MODERN	500– 1780 CE	R06 (Ships continued), R07 (Loom emerging)	<i>Maritime dominance → binary logic birth</i>	3
 ERA 5 — INDUSTRIAL REVOLUTION	1780– 1900 CE	R07 (Loom), R08 (Rail), R09 (Engine)	<i>Mechanisation → mass energy → acceleration</i>	5
 ERA 6 — MODERN ERA	1900– 2000 CE	R10 (AAA Triad), R11 (Orbit), R12 (Human Nodes)	<i>Triple mobility → celestial reach → biological network</i>	12
 ERA 7 — CONVERGENCE	2000– 2026+	R12 (Human Nodes), FC13 (Fractal Connector)	<i>Digital consciousness → system integration</i>	2

MILLIONS
OF YEARS

THOUSANDS
OF YEARS

CENTURIES

DECADES
& YEARS

ACCELERATION

IAAi

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THE DYAD TAXONOMY — BLOCK 617

The Five-Web Evolution of Human – Nonhuman Coupling

Excel Sheet 3:
Dyad Taxonomy
(Block 617)



PHYSICAL WEB

Dyad Type:	Mechanical Dyad
Non-Human Partner:	Inert machine
Cognitive Contribution:	None
Examples:	Bicycle, cart, train, crane
Governance:	Safety is physical (engineering standards)



BIOLOGICAL WEB

Dyad Type:	Animal Dyad
Non-Human Partner:	Horse, ox, dog, elephant
Cognitive Contribution:	Instinct only
Examples:	Horse-rider, ox-plough, falconry
Governance:	Safety is behavioural (training, domestication)



SOCIAL WEB

Dyad Type:	Human-Human Dyad
Non-Human Partner:	Another human
Cognitive Contribution:	Symmetric cognition
Examples:	Marriage, partnership, team dyads
Governance:	Safety is emotional (trust, norms, contracts)



DIGITAL WEB

Dyad Type:	Synthetic Dyad
Non-Human Partner:	AI system
Cognitive Contribution:	Asymmetric cognition
Examples:	Human + AI co-work, copilots, hybrid decision systems
Governance:	Safety is epistemic (beta-tested, Dearden's Law)



CONSCIOUSNESS WEB

Dyad Type:	Meta-Dyad
Non-Human Partner:	Hybrid cognition (human intention + machine intelligence)
Cognitive Contribution:	Joint problem-solving
Examples:	R12 Human Node, iAAi Dyad, hybrid cognition loops
Governance:	Safety is systemic (COUNTERFORCE, SES)

DYAD₆ — THE REVELATORY DYAD ($H \oplus \Omega$) — THE LOOM ITSELF

The environment within which the fabric is woven. Omnipresent. Unspoken. The frame that enables creation.

1

$$AD^2 = 16$$

As capability rises, resistance must rise faster.

2

$$C_{\text{true}} = \frac{A \times P}{\beta_{\text{tested}}}$$

Only the Synthetic Dyad requires epistemic safety.

3

SES Triad:

$$\text{Speed} = A \mid \text{Efficiency} = P \mid \text{Safety} = \beta_{\text{tested}}$$

SES is the control law of the Synthetic Dyad.



iAAi COUNTERFORCE v68 — Infrastructure Academy — *The Dyad is universal. The Synthetic Dyad is unprecedented. The Revelatory Dyad is omnipresent.*



ANCHOR EXAMPLE: INDUS VALLEY CIVILISATION

◆ — R03 (River) — 2600 BCE — East Mode — GREEN Zone — ◆

◇ CONTEXT ◇



Relay: R03 — River
(Cradles of Continuity)



Date: 2600 BCE



Zone: East



Mode: E (Eastern Continuity)



Initiator: Harappan Civilisation



Scholar: Sima Qian
(The Historian)



Surplus Type: Agricultural / Urban



Collapse Vector: Climate (drought)

◇ ISI CALCULATION ◇

INPUTS:

A (Ambition) = 7.5

P (Persistence) = 8.0

β (Drag) = 3.2

λ (Ethical) = 1.5

CALCULATION:

$A \times P = 60.0$

$R_{add} = \beta + \lambda = 4.7$

$R_{mult} = \beta \times \lambda = 4.8$

$C_{add} = 60.0 / 4.7$
= 12.77

$C_{mult} = 60.0 / 4.8$
= 12.50

RESULT:

$$\rho = \frac{C_{add}}{C_{mult}} = 1.62$$

◇ VERDICT ◇



SYSTEM IS ROBUST

Low β (minimal conflict in early period),
very low λ (pre-doctrinal era).
High persistence from continuous
river-fed agriculture. The system
endured 700+ years before
climate shift.

ANCHOR EXAMPLE: CHINA GRAND CANAL

◆ — R03 (River) — 486 BCE — East Mode — GREEN Zone — ◆

◇ CONTEXT ◇



Relay: R03 — River
(Cradles of Continuity)



Date: 486 BCE
(begun under King Fuchai of Wu)



Zone: East



Mode: E (Eastern Continuity)



Initiator: Kingdom of Wu /
successive Chinese dynasties



Scholar: Confucius / Sima Qian



Surplus Type: Agricultural / Trade



Collapse Vector: Governance fragmentation

◇ ISI CALCULATION ◇

INPUTS:

A (Ambition) = 7.0

P (Persistence) = 9.0

β (Drag) = 3.5

λ (Ethical) = 1.2

CALCULATION:

$A \times P = 63.0$

$R_{add} = \beta + \lambda = 4.7$

$R_{mult} = \beta \times \lambda = 4.2$

$C_{add} = 63.0 / 4.7$
= 13.40

$C_{mult} = 63.0 / 4.2$
= 15.00

RESULT:

$$\rho = \frac{C_{add}}{C_{mult}} = 1.87$$

◇ VERDICT ◇



SYSTEM IS ROBUST

Exceptional persistence (P=9.0) — the canal was maintained and extended across 2,500 years by successive dynasties. Low ethical friction in a Confucian governance system.

Deep continuity mode. The longest continuously operated infrastructure project in human history.

ANCHOR EXAMPLE: THE MARSHALL PLAN

R05 (Roads) / R09 (Engine) — 1948 CE — West Mode — AMBER Zone

CONTEXT



Relay: R05 (Roads) / R09 (Engine)
— cross-relay



Date: 1948 CE



Zone: West



Mode: W (Western Projection)



Initiator: United States /
George C. Marshall



Scholar: Aristotle
(Logic of reconstruction)



Surplus Type: Industrial / Financial



Collapse Vector: Cost (massive capital
outlay) + Conflict
(Cold War pressure)

ISI CALCULATION

INPUTS:

A (Ambition)	= 8.5
P (Persistence)	= 7.0
β (Drag)	= 5.0
λ (Ethical)	= 2.5

CALCULATION:

$A \times P = 59.5$	$C_{add} = 59.5 / 7.5$ = 7.93
$R_{add} = \beta + \lambda = 7.5$	$C_{mult} = 59.5 / 12.5$ = 4.76
$R_{mult} = \beta \times \lambda = 12.5$	

RESULT:

$$\rho = \frac{C_{add}}{C_{mult}} = 2.22$$

VERDICT



SYSTEM IS SENSITIVE

High β from post-war devastation and Cold War geopolitical drag. Moderate λ from ethical tensions (selective aid, political conditions). High ambition succeeded but required sustained political will. The system worked — but was fragile to withdrawal.



ANCHOR EXAMPLE: CERN / WORLD WIDE WEB

R11 (Orbit) — 1989 — Unified Mode — GREEN (Edge of AMBER)

◇ CONTEXT ◇

	Relay: R11 — Orbit (The Celestial Relay)
	Date: 1989
	Zone: Unified (International)
	Mode: U (Unified / Collaborative)
	Initiator: Tim Berners-Lee / CERN Consortium
	Scholar: Berners-Lee (The Web — Tier 2 Diagnostician)
	Surplus Type: Information / Knowledge
	Collapse Vector: Governance deficit (open vs controlled)

◇ ISI CALCULATION ◇

INPUTS:

A (Ambition)	=	9.0
P (Persistence)	=	7.5
β (Drag)	=	4.5
λ (Ethical)	=	2.0

CALCULATION:

$A \times P = 67.5$	$C_{add} = 67.5 / 6.5$ = 10.38
$R_{add} = \beta + \lambda = 6.5$	$C_{mult} = 67.5 / 9.0$ = 7.50
$R_{mult} = \beta \times \lambda = 9.0$	

RESULT:

$$\rho = \frac{C_{add}}{C_{mult}} = 1.97$$

◇ VERDICT ◇



SYSTEM ROBUST — BUT AT THRESHOLD

Extremely high ambition (global information network).
Moderate β from Cold War legacy and
institutional complexity. $\lambda = 2.0$ reflects the
ethical tension between open access and control.
 $\rho = 1.97$ sits just below the 2.0 AMBER
boundary — the system works but needs
governance to stay green.

FC13: THE FRACTAL CONNECTOR – SUN TZU CHAPTER 13

Why Thirteen — The Intelligence Layer Binding All 12 Relays

THE SUN TZU PARALLEL



The Art of War:
13 Chapters



Chapters 1–12:
Tactical operations
(the 12 relays)



Chapter 13:
“The Use of Spies” —
the intelligence layer



The invisible web that
makes all other
operations possible



Without Chapter 13,
the first 12 are blind



THE PLAYING CARDS ANALOGY



13 cards per suit



12 = numbered cards
(the relays)



The 13th completes
the suit



Enables the deal
to begin again



The recursive loop —
fractal by nature



The 13th Composition:
From Calories to Consciousness —
A Civil Engineer’s Perspective and Guide —
An Infrastructure Odyssey (2026)

Per Arya Ad Astra —
Through Noble Effort to the Stars



THE SEARCH FOR THE 8TH SCHOLAR

The 56-Year Generational Relay (1989–2045+)



GILBERT STRAND
The System Architect



BUILD → TRANSFER → ELEVATE



DEARDEN SYSTEMATISES
The Living Framework



**APPLICATION
ASCENSION
LEGACY**



TIER 5: THE APPLIED
1989–2045+
56 Years of Generational Relay



THE 7TH VOICE SPEAKS
The System Builder completes the framework
12 Relays. 37 Examples. One Unified Science.



THE 8TH SCHOLAR EMERGES
Who carries the baton next?
Per Arya Ad Astra — Through Struggle to the Stars

◇ *From Calories to Consciousness: A Civil Engineer's Perspective and Guide — An Infrastructure Odyssey* ◇

Ir. Nigel T. Dearden BEng(Hons) CEng MICE MHKIE MCIWEM C.WEM RPE(CSB)



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